

ARTIFICIAL INTELLIGENCE AND HEALTHCARE ECONOMICS

Evidence, Value, Governance, and Sustainability

START HERE

Digital Resource Companion V1.0

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PRACTICAL RECORDS · WORKBOOKS · WORKED EXAMPLES · ANALYTICS · WEB COMPANION

Purpose of the companion

The Digital Resource Companion is the applied extension of Artificial Intelligence and Healthcare Economics: Evidence, Value, Governance, and Sustainability. The book contains the durable conceptual, economic, methodological, and governance argument. The companion provides editable institutional records, structured workbooks, worked examples, decision pathways, reference guides, process maps, and reproducible analytical resources that support application and local adaptation.

The companion does not replace the reasoning developed in the book. It provides a structured way to apply that reasoning to a clearly specified jurisdiction, organization, population, workflow, intervention, comparator, technology version, evidence base, and accountable decision process.

Choose the appropriate entry point

Entry point	Best suited to	What it provides
GitHub Pages Web Companion	Most users beginning online	Guided navigation by chapter, decision stage, form, workbook, worked example, or analytical method.
Reader and Practitioner Edition	Readers, practitioners, managers, analysts, educators, and institutional teams	A complete, accessible offline collection with the Web Companion, forms, workbooks, cases, guides, selected analytics, catalogues, and printable companion.
Core Tools Pack	Teams that primarily need the principal institutional documents	The consolidated Core Resource Forms, consolidated Core Resource Workbook, practical guidance, licence, disclaimer, and package manifest.
Canonical Source	Researchers, developers, repository maintainers, advanced analysts, educators, and institutions	The complete authoritative source collection, including editable, computational, machine-readable, process, publication, and website materials.
Printable Digital Resource Companion	Readers who prefer a coherent readable reference	A book-like PDF and Word volume explaining the companion architecture, resources, cases, methods, and recommended use.

Recommended starting point. Most readers should begin with the Web Companion or the Reader and Practitioner Edition. Use the Core Tools Pack when the immediate need is a compact set of institutional forms and workbooks. Use the Canonical Source when maintaining, extending, reproducing, auditing, or republishing resources.

V1.0 resource scope

Resource class	V1.0 scope
Book structure	5 Parts and 15 chapters
Principal book figures	31
Principal book tables	46
Core institutional records	24
Worked examples	20, plus AIHE-CASE00 as the common case-analysis method
Python analytical modules	40
Catalogued companion resources	146

Stable identifiers should be used in preference to printed page numbers. Examples include AIHE-D02 for the lifecycle decision-state record, AIHE-CASE15 for the remote-diagnostic worked example, and AIHE-PY38 for the related circularity-adjusted analysis.

How the companion is organized

Component	Role
Chapter-aligned resources	Canonical records, guides, analytical modules, and process material aligned principally with each book chapter.
Decision pathways	Problem-first routes that help users choose the smallest sufficient set of resources for a particular institutional question.
Core forms	Narrative records for committee review, justification, conditions, accountability, sign-off, and decision provenance.
Core workbooks	Structured spreadsheets for registers, costing, comparison, monitoring, calculations, and repeated review.
Worked examples	Fictional scenarios and synthetic datasets showing how book concepts and companion tools can be applied.
Python analytics	Reproducible modules for data quality, evaluation, modelling, uncertainty, procurement, monitoring, asset life, and sustainable value.
Catalogues	Machine-readable and spreadsheet indexes for resources, figures, tables, cases, Python modules, and relationships.
Web Companion	Browser-based access to current forms, process views, worked examples, catalogues, and decision pathways.

The five core resource families

Family	Scope	Core record IDs
A — Decision framing, participation, readiness, and data	Decision specification, stakeholders, institutional readiness, and data fitness.	AIHE-A01 to AIHE-A04
B — Economics, affordability, benefits, and modelling	Economic specification, lifecycle cost, capacity conversion, HTA, modelling, uncertainty, and reproducibility.	AIHE-B01 to AIHE-B05
C — Quality, risk, accountability, and communication	Quality control, corrective action, institutional risk, human oversight, and public communication.	AIHE-C01 to AIHE-C05
D — Procurement and lifecycle governance	Vendor requirements, decision states, monitoring, incidents, material change, renewal, replacement, and retirement.	AIHE-D01 to AIHE-D05
E — Circularity, assets, and sustainable value	Circular-economy assessment, circular procurement, supported asset life,	AIHE-E01 to AIHE-E05

Family	Scope	Core record IDs
	circularity-adjusted evaluation, and disposition monitoring.	

The resource families are not mandatory sequences. A decision should use only the records needed to make assumptions, evidence, responsibilities, safeguards, resource consequences, and lifecycle conditions sufficiently explicit.

Recommended institutional sequence

1. Define the service problem, affected population, setting, intended use, accountable decision-maker, and realistic comparator.
2. Read the aligned book chapter and identify the smallest sufficient companion resource set.
3. Review the corresponding worked example and distinguish its synthetic assumptions from evidence needed locally.
4. Create a governed local copy of each selected record or workbook.
5. Replace illustrative content with authorized, versioned local evidence and preserve important unknowns or disagreements.
6. Complete multidisciplinary review proportionate to consequence, uncertainty, scale, reversibility, and institutional dependence.
7. Record the current decision state, conditions, mandatory safeguards, monitoring thresholds, transition triggers, accountable authority, next review date, and credible exit route.
8. Reconsider the decision after material change, new evidence, an incident, changed costs, renewal, restriction, replacement, or retirement.

Decision lifecycle at a glance

Stage	Principal question	Typical companion records
1. Define the decision	What problem is being addressed, compared with what, for whom, and under whose authority?	AIHE-A01; AIHE-A02
2. Assess readiness and data	Are the required data, infrastructure, workforce, workflow, governance, and service capabilities adequate?	AIHE-A03; AIHE-A04
3. Establish evidence and quality requirements	What evidence, validation, quality control, thresholds, and assurance are proportionate to use?	AIHE-C01; AIHE-C02
4. Model cost, value, affordability, and HTA	What are the lifecycle resources, financial consequences, outcomes, budget effects, and distribution?	AIHE-B01 to AIHE-B04
5. Model uncertainty and research value	Which model structure is credible, and could further evidence change the decision?	AIHE-B05
6. Assess governance and institutional risk	Are decision rights, equity, privacy, cybersecurity, continuity, trust, and communication adequate?	AIHE-C03 to AIHE-C05
7. Procure and contract	Can supplier evidence, pricing, updates, audit rights, portability, remedies, and exit be governed?	AIHE-D01
8. Implement, monitor, and respond	Does use remain within approved performance, safety, workload, equity, cost, and governance conditions?	AIHE-D02 to AIHE-D04
9. Assess circular and sustainable value	Can value be preserved while controlling resource demand, supported life, environmental effects, and disposition?	AIHE-E01 to AIHE-E05
10. Renew or end authorization	Is continuing use still preferable to redesign, restriction, replacement, or retirement?	AIHE-D02; AIHE-D05

Using forms and workbooks together

Word forms	Excel workbooks
Best for narrative justification, committee discussion, conditions, responsibilities, dissent, authorization, and sign-off.	Best for structured data entry, costing, registers, comparisons, calculations, monitoring, and repeated review.
Use the consolidated volume when a committee needs	Use the consolidated workbook when a project requires

Word forms	Excel workbooks
several records in one controlled document.	several structured records in one file.
Use an individual form when only one institutional record is needed.	Use an individual workbook when a single analytical or tracking task is sufficient.
Preserve the resource ID, V1.0, evidence cut-off, intervention version, owner, and local-modification record.	Preserve the same controls and document data definitions, formula assumptions, and evidence sources.

A form and workbook may be used together. The form provides the narrative institutional record; the workbook provides the structured evidence supporting that record. Neither format should become an uncontrolled substitute for the current decision dossier.

Worked examples and computational resources

Worked examples illustrate the level of specificity needed to connect a service problem with evidence, economics, governance, implementation, monitoring, and lifecycle decisions. The scenarios and datasets are synthetic unless explicitly stated otherwise. They should not be cited as empirical evidence for a local decision.

The V1.0 collection contains twenty worked examples, together with AIHE-CASE00 as a common method for analysing a case. Supplementary examples address financial recognition and budget incidence, the expected value of additional evidence, upstream model-provider change, rebound and burden transfer, and the circularity-adjusted evaluation of an AI-assisted remote diagnostic pathway.

The Canonical Source contains forty Python analytical modules. Smaller distributions carry a selected set of high-value modules while retaining the full catalogue. Computational results require methodologically appropriate inputs, verification, validation, documented assumptions, and interpretation by people with relevant expertise.

Version control and reproducibility

- Record the stable resource, case, or module identifier and V1.0.
- Record the date accessed, evidence cut-off, intervention or model version, and local modifications.
- Preserve the exact input data, code, environment, assumptions, and outputs used in consequential analysis.
- Do not rely on printed page numbers as persistent companion identifiers.
- Review regulatory, policy, standards, software, and supplier information for currency before institutional reliance.
- After material change, determine explicitly which prior evidence and authorizations remain applicable.

Responsible use

The companion supports education, structured analysis, research, institutional deliberation, and local adaptation. It is not a validated clinical instrument and does not replace legal, regulatory, clinical, ethical, privacy, cybersecurity, procurement, accreditation, financial, or health technology assessment review.

- Do not enter identifiable patient information, credentials, private keys, confidential contracts, or restricted institutional information into public browser demonstrations or uncontrolled files.
- Do not interpret a completed form, score, or model result as automatic authorization.
- Do not allow favourable aggregate results to override a mandatory safeguard.
- Use accountable human authority for every consequential decision-state transition.
- Maintain a credible route for restriction, rollback, replacement, retirement, and responsible closure.

Quick links

Resource	Address
Web Companion	https://ai-healthcare-economics.github.io/digital-resource-companion/
Canonical GitHub repository	https://github.com/ai-healthcare-economics/digital-resource-companion
GitHub Releases	https://github.com/ai-healthcare-economics/digital-resource-companion/releases
Zenodo V1.0 DOI	https://doi.org/10.5281/zenodo.21753927

For scholarly or institutional use, cite the version-specific DOI and record the exact companion resource identifier and any local changes.